23 August 2026

To the Editors, *IEEE Access*

Dear Editors,

We submit for your consideration the article **"Surrogate Selection for Reinforcement-Learning HVAC Control: Step Size, Not Predictive Accuracy, Predicts Control Utility"** together with its supplementary material.

What the article reports. Reinforcement-learning controllers for building HVAC systems are trained against fast learned surrogates, and the surrogate is selected by predictive error. On the BOPTEST benchmark we find that criterion to be not merely uninformative but inverted: the least accurate surrogate in the study is the only one that trains a usable controller, while the most accurate one trains a controller that leaves the comfort band for more than 77% of the horizon on every random seed. We identify the property that does predict training utility, the magnitude of the surrogate's per-step increment, and isolate it with a rescaling control that holds response-surface roughness fixed while changing only the integration step. We then give an architecture, role-separated surrogate training, that acts on the finding, and a receding-horizon MPC baseline showing that a planner reproduces the same inverted ordering, so the effect belongs to optimising through the surrogate rather than to reinforcement learning specifically.

Why it is verifiable. Every experiment runs on BOPTEST, the open IBPSA-derived benchmark, rather than on a private building model, so the results can be reproduced and contested by other groups. Five hypotheses were written down with their pass criteria before the corresponding runs; three are falsified, including two of our own predictions, and all are reported as they came out. The central comparisons are replicated across random seeds, and the censor-weight sweep is replicated at every swept setting. One planned experiment was executed and then invalidated by a defect we found in our own planner; that run is recorded in the project artifacts rather than discarded, and the corrected run supersedes it.

Fit to IEEE Access. The contribution is methodological, about how to choose a training environment for a learning controller, demonstrated on a building-control case study. It sits between machine learning and building systems, and its central result is a negative one about a criterion the field uses by default. IEEE Access states that it welcomes multidisciplinary and applications-oriented work and negative results that do not fit the traditional journals, which is the reason we bring the article here.

Disclosures. The work is original and is not under consideration elsewhere. Earlier results from this study were presented as a conference talk at the WCCM–ECCOMAS Congress 2026 (Munich, 19–24 July 2026); the present article substantially extends that material and shares no text with it. All authors have approved the submission and declare no competing interests. Author biographies and photographs are not included with this submission and can be supplied if the article proceeds. The project codebase is not publicly released at this stage; it is available to the editors and reviewers on request, together with the derived artifacts, from the corresponding author.

Thank you for considering the article.

Yours sincerely,

Almaz Sapargali, on behalf of all authors
Al-Farabi Kazakh National University and LLP Digit Alem, Almaty, Kazakhstan
almaz.sapargali2001@gmail.com